Norway

SLEIPNER
Volve F
F-15

F-15D

Plan: 15/9-F-15 D revp8

Standard Survey Report

11 april, 2018

Statoil
Survey Report

Company:	Norway	**Local Co-ordinate Reference:**	Site Volve F
Project:	SLEIPNER	**TVD Reference:**	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	**MD Reference:**	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	**North Reference:**	Grid
Wellbore:	F-15D	**Survey Calculation Method:**	Minimum Curvature
Design:	15/9-F-15 D revp8	**Database:**	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	**System Datum:**	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9				
Site Position:		**Northing:**	6,478,563.53 m	**Latitude:**	58° 26' 29.8068 N
From:	Map	**Easting:**	435,050.02 m	**Longitude:**	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	**Slot Radius:**	13.200 in	**Grid Convergence:**	-0.95 °

Well	F-15					
Well Position	**+N/-S**	-3.17 m	**Northing:**	6,478,560.36 m	**Latitude:**	58° 26' 29.7063 N
	+E/-W	3.53 m	**Easting:**	435,053.55 m	**Longitude:**	1° 53' 15.1494 E
Position Uncertainty		0.00 m	**Wellhead Depth:**	91.00 m	**Water Depth:**	91.00 m

Wellbore F-15D

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	ɜNETICREFERENCE	25.06.2013	-1.17	71.62	50,516

Design 15/9-F-15 D revp8

Audit Notes:

Version:	**Phase:** PLAN	**Tie On Depth:**	1,364.50

Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-3.17	3.53	84.64

Survey Tool Program **Date** 19.01.2016

From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
153.40	2,286.70	12 1/4" F-15 Drop Gyro (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD
2,329.90	3,925.90	8 1/2" F-15 Drop Gyro Revised (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD
3,963.80	1,364.50	F15 8 1/2" MWD Surveys (F-15)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
1,364.50	2,613.00	15/9-F-15 D revp8 (F-15D)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
2,613.00	3,298.00	15/9-F-15 D revp8 (F-15D)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
3,298.00	4,701.54	15/9-F-15 D revp8 (F-15D)	Magn, IFR, non-mag, dual in	Magnetic Tools (MWD, EMS)

Planned Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,364.50	24.71	243.03	1,336.81	-182.34	-74.67	-94.59	0.000	0.000	0.000
1,368.40	24.71	243.03	1,340.35	-183.08	-76.12	-96.10	0.000	0.000	0.000
1,378.40	24.71	243.03	1,349.44	-184.98	-79.85	-99.99	0.000	0.000	0.000
1,478.40	18.92	265.28	1,442.39	-195.82	-114.72	-135.72	3.000	-1.736	6.675
1,488.40	18.92	265.28	1,451.85	-196.09	-117.96	-138.97	0.000	0.000	0.000
1,935.74	30.00	3.64	1,872.08	-86.57	-185.55	-196.04	2.500	0.743	6.596

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Planned Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,384.03	30.00	3.64	2,260.33	137.10	-171.34	-161.01	0.000	0.000	0.000
2,632.80	32.78	43.71	2,474.97	249.06	-120.29	-99.74	2.500	0.336	4.833
2,912.51	59.77	46.05	2,666.52	390.29	21.68	54.80	2.900	2.895	0.251
2,924.79	59.77	46.05	2,672.70	397.66	29.32	63.09	0.000	0.000	0.000
3,511.97	74.24	106.40	2,920.90	502.65	524.16	565.57	2.900	0.739	3.083
3,780.69	74.24	106.40	2,993.90	429.63	772.25	805.76	0.000	0.000	0.000
3,821.75	74.16	109.96	3,005.08	417.30	809.78	841.97	2.500	-0.056	2.598
4,143.77	74.16	109.96	3,092.98	311.56	1,100.97	1,122.03	0.000	0.000	0.000
4,192.14	77.30	107.35	3,104.90	296.58	1,145.38	1,164.84	2.500	1.950	-1.615
4,501.54	77.30	107.35	3,172.90	206.55	1,433.48	1,443.28	0.000	0.000	0.000
4,701.54	77.30	107.35	3,216.86	148.35	1,619.71	1,623.26	0.000	0.000	0.000

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
F-15 E IOR Intra Hugin - plan hits target center - Point	0.00	0.00	2,920.90	502.65	524.16	6,479,066.00	435,574.00	58° 26' 46.3289 N	1° 53' 46.7236 E
F-15 E IOR Guide#1 (18 - plan hits target center - Point	0.00	0.00	2,993.90	429.63	772.25	6,478,993.00	435,822.00	58° 26' 44.1005 N	1° 54' 2.0886 E
F-15 E IOR Guide#2 (18 - plan hits target center - Point	0.00	360.00	3,104.90	296.58	1,145.38	6,478,860.00	436,195.00	58° 26' 39.9978 N	1° 54' 25.2204 E
F-15 E IOR Guide#3 (18 - plan hits target center - Point	0.00	360.00	3,172.90	206.55	1,433.48	6,478,770.00	436,483.00	58° 26' 37.2395 N	1° 54' 43.0671 E

Casing Points

Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
220.60	220.60	30"	30.000	36.000
1,368.40	1,340.35	20" Surface Casing	20.000	26.000
2,613.00	2,458.26	13 3/8" Production Casing	13.375	17.500
3,298.00	2,847.83	9 5/8" Production Casing	9.625	12.250
4,701.54	3,216.86	7" Production Liner	7.000	8.500

Checked By: ____________ Approved By: ____________ Date: ____________